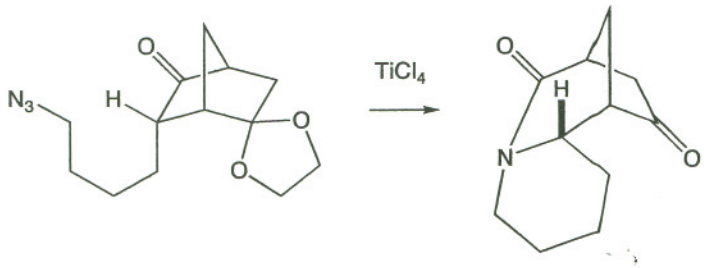
ORGANIC CUME
APRIL 4, 2003

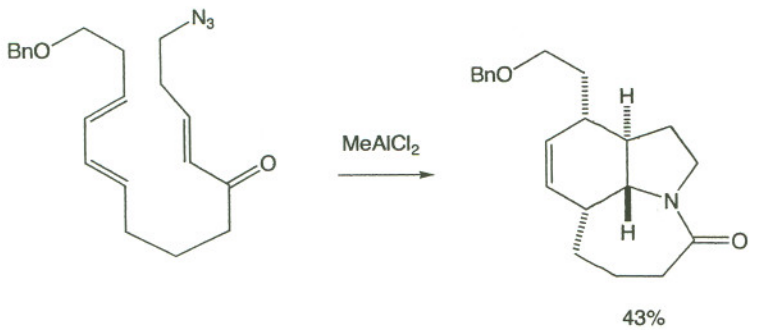
References:

Aubé et al *J. Am. Chem. Soc.* **2002**, *124*, 9974.
Aubé and Golden, *Angew. Chem. Int. Ed.* **2002**, *41*, 4316
Aubé et al, *Org. Lett.* **2002**, *4*, 2577.

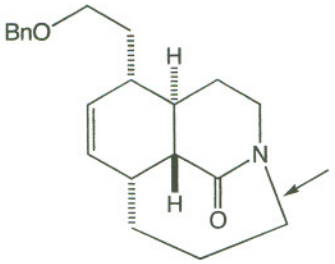
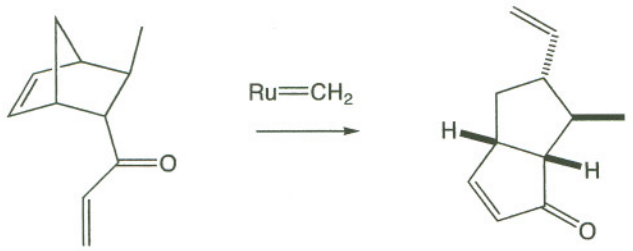
1. Draw the complete step-by-step electron pushing mechanism for the rearrangement step in the following transformation (ignore the acetal deprotection step).



2. Draw the complete step-by-step electron pushing mechanism for the following transformation.



3. The amide below is another product of the above reaction. It undergoes some unusual chemical reactions such as reductive cleavage of the indicated C-N bond upon treatment with Pd/C and H_2. What is responsible for the unusual chemical behavior of the amide?



4. Draw a complete step-by-step mechanism for the following reaction. Use $Ru{=}CH_2$ in place of the full catalyst structure.